

$$\dot{n} = V \sin(\chi), \quad (1)$$

$$\dot{\xi} = \gamma - \frac{d\phi_{\text{ref}}}{ds} \frac{ds}{dt} \quad (2)$$

$$= \gamma - \kappa \dot{s}, \quad (3)$$

$$\dot{s} = \frac{V \cos(\chi)}{1 - \kappa n}. \quad (4)$$

$$\frac{dt}{ds} = \frac{1 - \kappa n}{V \cos(\chi)} =: S_f. \quad (5)$$

$$-\mathcal{N}_r + \left(\frac{w_t}{2} + w_s \right) \leq n \leq \mathcal{N}_l - \left(\frac{w_t}{2} + w_s \right), \quad (6)$$

$$a_x = \frac{1}{m_v} \left[F_{xf} \cos(\delta) - F_{yf} \sin(\delta) + F_{xr} - F_{\text{drag}} \right], \quad (7)$$

$$a_y = \frac{1}{m_v} \left[F_{yf} \cos(\delta) + F_{xf} \sin(\delta) + F_{yr} \right], \quad (8)$$

$$a_t = \frac{1}{m_v} \left[F_{xf} \cos(\delta - \beta) - F_{yf} \sin(\delta - \beta) + F_{xr} \cos(\beta) + F_{yr} \sin(\beta) - F_{\text{drag}} \cos(\beta) \right], \quad (9)$$

$$a_n = \frac{1}{m_v} \left[F_{xf} \sin(\delta - \beta) + F_{yf} \cos(\delta - \beta) - F_{xr} \sin(\beta) + F_{yr} \cos(\beta) + F_{\text{drag}} \sin(\beta) \right], \quad (10)$$

$$\begin{aligned} M_z &= l_f F_{yf} \cos(\delta) + l_f F_{xf} \sin(\delta) - l_r F_{yr} \\ &\quad + \frac{w_t}{2} \left[(F_{yfl} - F_{yfr}) \sin(\delta) + (F_{xfr} - F_{xfl}) \cos(\delta) + (F_{xrr} - F_{xrl}) \right], \end{aligned} \quad (11)$$

$$\dot{V} = a_t, \quad (12)$$

$$\dot{\beta} = \dot{\phi} - \dot{\psi} = \frac{a_n}{V} - \gamma, \quad (13)$$

$$\dot{\gamma} = \frac{M_z}{I_z}, \quad (14)$$

$$F_{zfl} = \frac{1}{2} m_v \left(\frac{l_r}{l} g - \frac{h_c}{l} \bar{a}_x \right) - m_v \left(\frac{l_r}{l} g - \frac{h_c}{l} \bar{a}_x \right) \frac{h_c}{w_t} \frac{\bar{a}_y}{g} - \frac{F_{\text{lift}}}{4}, \quad (15)$$

$$F_{zfr} = \frac{1}{2} m_v \left(\frac{l_r}{l} g - \frac{h_c}{l} \bar{a}_x \right) + m_v \left(\frac{l_r}{l} g - \frac{h_c}{l} \bar{a}_x \right) \frac{h_c}{w_t} \frac{\bar{a}_y}{g} - \frac{F_{\text{lift}}}{4}, \quad (16)$$

$$F_{zrl} = \frac{1}{2} m_v \left(\frac{l_f}{l} g + \frac{h_c}{l} \bar{a}_x \right) - m_v \left(\frac{l_f}{l} g + \frac{h_c}{l} \bar{a}_x \right) \frac{h_c}{w_t} \frac{\bar{a}_y}{g} - \frac{F_{\text{lift}}}{4}, \quad (17)$$

$$F_{zrr} = \frac{1}{2} m_v \left(\frac{l_f}{l} g + \frac{h_c}{l} \bar{a}_x \right) + m_v \left(\frac{l_f}{l} g + \frac{h_c}{l} \bar{a}_x \right) \frac{h_c}{w_t} \frac{\bar{a}_y}{g} - \frac{F_{\text{lift}}}{4}, \quad (18)$$

$$\bar{a}_x - a_x = 0, \quad (19)$$

$$\bar{a}_y - a_y = 0. \quad (20)$$

$$\dot{\bar{a}}_x = \frac{1}{\tau_{\text{acc}}} (a_x - \bar{a}_x) \quad (21)$$

$$\dot{\bar{a}}_y = \frac{1}{\tau_{\text{acc}}} (a_y - \bar{a}_y) \quad (22)$$

$$T_f = k_t T_t + k_b T_b, \quad (23)$$

$$T_r = (1 - k_t) T_t + (1 - k_b) T_b, \quad (24)$$

$$T_{fl} = \frac{T_f}{2} \left(1 - \frac{\Delta F_{zf}}{F_{zf}} \right), \quad T_{fr} = \frac{T_f}{2} \left(1 + \frac{\Delta F_{zf}}{F_{zf}} \right), \quad (25)$$

$$T_{rl} = \frac{T_r}{2} \left(1 - \frac{\Delta F_{zr}}{F_{zr}} \right), \quad T_{rr} = \frac{T_r}{2} \left(1 + \frac{\Delta F_{zr}}{F_{zr}} \right), \quad (26)$$

ative vertical loads are calculated as

$$\Delta F_{zf} = \frac{1}{2} (F_{zfr} - F_{zfl}) = m_v \left(\frac{l_r}{l} g - \frac{h_c}{l} \bar{a}_x \right) \frac{h_c}{w_t} \frac{\bar{a}_y}{g}, \quad (27)$$

$$\Delta F_{zr} = \frac{1}{2} (F_{zrr} - F_{zrl}) = m_v \left(\frac{l_f}{l} g + \frac{h_c}{l} \bar{a}_x \right) \frac{h_c}{w_t} \frac{\bar{a}_y}{g}. \quad (28)$$

$$\dot{\omega}_i = \frac{1}{I_w} (T_i - r_w F_{xi}), \quad i \in \{fl, fr, rl, rr\}, \quad (29)$$

$$V_{xfl} = (V_x - \frac{w_t}{2}\gamma) \cos(\delta) + (V_y + l_f\gamma) \sin(\delta), \quad (30)$$

$$V_{xfr} = (V_x + \frac{w_t}{2}\gamma) \cos(\delta) + (V_y + l_f\gamma) \sin(\delta), \quad (31)$$

$$V_{xrl} = V_x - \frac{w_r}{2}\gamma, \quad (32)$$

$$V_{xrr} = V_x + \frac{w_r}{2}\gamma, \quad (33)$$

ocities of the tire trajectories are given by

$$V_{yfl} = (V_y + l_f\gamma) \cos(\delta) - (V_x - \frac{w_t}{2}\gamma) \sin(\delta), \quad (34)$$

$$V_{yfr} = (V_y + l_f\gamma) \cos(\delta) - (V_x + \frac{w_t}{2}\gamma) \sin(\delta), \quad (35)$$

$$V_{yrl} = V_y - l_r\gamma, \quad (36)$$

$$V_{yrr} = V_y + l_r\gamma. \quad (37)$$

$$\lambda_i = \frac{r_w\omega_i - V_{xi}}{V_{xi}}, \quad (38)$$

$$\alpha_i = \tan^{-1}\left(\frac{V_{yi}}{V_{xi}}\right). \quad (39)$$

$$F_{x0}(\lambda, F_z, \mu) = \frac{\mu}{\mu_0} D_x \sin\left[C_x \tan^{-1}(B_x \lambda)\right], \quad (40)$$

where μ_0 is the road friction coefficient during tire testing, and B_x , C_x , and D_x are fitting coefficients. Since the peak factor D_x varies linearly with the vertical tire force, it can be modeled as a first-order function of F_z with a linear term d_1 and a constant term d_2 ,

$$D_x = d_{1x}F_z + d_{2x}. \quad (41)$$

The simplified MF for the lateral tire forces under pure sideslip conditions is shown in the right plot of Figure 3, and it is expressed by the following equation:

$$F_{y0}(\alpha, F_z, \mu) = -\frac{\mu}{\mu_0} D_y \sin\left[C_y \tan^{-1}(B_y \alpha)\right], \quad (42)$$

where

$$D_y = d_{1y}F_z + d_{2y}. \quad (43)$$

$$\sigma_i = \sqrt{\lambda_i^2 + \tan^2(\alpha_i)}, \quad i \in \{fl, fr, rl, rr\}. \quad (44)$$

And the combined slip longitudinal and lateral tire forces of are defined by

$$F_{xi} = \frac{\lambda_i}{\sigma_i} F_{x0}(\sigma_i, F_{zi}, \mu), \quad F_{yi} = \frac{\tan(\alpha_i)}{\sigma_i} F_{y0}(\sigma_i, F_{zi}, \mu), \quad i \in \{fl, fr, rl, rr\}. \quad (45)$$